

Assignments

assignment 10

sb-prob1611.problem

Due date: Sun Apr 5 11:59:59 pm 2026 (EDT)

Transverse waves with a speed of 46.5m/s are to be produced in a taut string. A 5.47m length of string with a total mass of 0.0562kg is used. What is required tension?

Tries 0/10

sb-prob1629a.problem

A sinusoidal wave is described by the equation $D(x,t)=(0.690\text{m})\sin(0.610x-0.482t)$, where x is in meters and t is in seconds. What is the amplitude of the wave?

Tries 0/10

What is the angular frequency of the wave?

Tries 0/10

What is the angular wave number?

Tries 0/10

What is the wavelength?

Tries 0/10

What is the speed of the wave?

Tries 0/10

sb-prob1632a.problem

A transverse sinusoidal wave on a string has a period $T = 25.2\text{ms}$ and travels in the negative x direction with a speed of 30.0m/s. At $t = 0$, a particle on the string at $x = 0$ has a displacement of 2.00cm and travels downward with a speed of 1.80m/s. What is the amplitude of the wave?

Tries 0/10

What is the initial phase constant assuming that $y(x,t)$ is a sine function? Enter a value between 0 and 360° .

Tries 0/10

What is the maximum transverse speed of the string?

Tries 0/10

sb-prob1642b.problem

It is found that a 6.01m segment of a long string contains four complete waves and has a mass of 171g. The string is vibrating sinusoidally with a frequency of 46.0Hz and a peak-to-valley displacement of 14.8cm. (The "peak-to-valley" distance is the vertical distance from the farthest positive displacement to the farthest negative displacement.) Determine the power being supplied to the string.

Tries 0/10

kn-prob2008.problem

(2) The wave speed on a string is 148.0 m/s when the tension is 71.0 N. What tension will give a speed 190.0 m/s?

Tries 0/10

kn-prob2062a.problem

(8) A string with linear density 2.95 g/m is stretched along the positive x -axis with tension 22.7 N. One end of the string, at $x = 0.00$ m, is tied to a hook that oscillates up and down at a frequency of 139.0Hz with a maximum displacement of 0.207 mm. At $t = 0.00$ s, the hook is at its lowest point.

(a) What is the wave speed on the string?

Tries 0/10

(b) What is the wavelength?

Tries 0/10

(c) What is the amplitude of the wave?

Tries 0/10